

The SAS System

The Mixed Procedure

Model Information	
Data Set	WORK.NEW2
Dependent Variable	Amyloid
Covariance Structure	Unstructured
Subject Effect	studyid
Estimation Method	REML
Residual Variance Method	None
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Between-Within

Class Level Information		
Class	Levels	Values
studyid	157	IAM_1001 IAM_1002 IAM_1003 IAM_1004 IAM_1006 IAM_1008 IAM_1009 IAM_1010 IAM_1011 IAM_1012 IAM_1013 IAM_1014 IAM_1015 IAM_1016 IAM_1017 IAM_1018 IAM_1019 IAM_1020 IAM_1021 IAM_1022 IAM_1023 IAM_1025 IAM_1026 IAM_1027 IAM_1028 IAM_1029 IAM_1030 IAM_1031 IAM_1032 IAM_1033 IAM_1034 IAM_1035 IAM_1036 IAM_1037 IAM_1039 IAM_1040 IAM_1041 IAM_1042 IAM_1043 IAM_1044 IAM_1045 IAM_1046 IAM_1047 IAM_1048 IAM_1049 IAM_1050 IAM_1051 IAM_1052 IAM_1053 IAM_1054 IAM_1055 IAM_1056 IAM_1057 IAM_1058 IAM_1059 IAM_1062 IAM_1063 IAM_1064 IAM_1065 IAM_1066 IAM_1067 IAM_1068 IAM_1069 IAM_1071 IAM_1072 IAM_1073 IAM_1074 IAM_1075 IAM_1076 IAM_1077 IAM_1078 IAM_1079 IAM_1080 IAM_1081 IAM_1082 IAM_1083 IAM_1084 IAM_1085 IAM_1086 IAM_1087 IAM_1088 IAM_1089 IAM_1090 IAM_1091 IAM_1092 IAM_1093 IAM_1094 IAM_1095 IAM_1096 IAM_1097 IAM_1098 IAM_1099 IAM_1101 IAM_1102 IAM_1103 IAM_1104 IAM_1105 IAM_1106 IAM_1107 IAM_1108 IAM_1109 IAM_1110 IAM_1111 IAM_1112 IAM_1113 IAM_1114 IAM_1115 IAM_1116 IAM_1117 IAM_1118 IAM_1119 IAM_1120 IAM_1121 IAM_1122 IAM_1123 IAM_1124 IAM_1125 IAM_1126 IAM_1127 IAM_1128 IAM_1129 IAM_1130 IAM_1131 IAM_1132 IAM_1133 IAM_1134 IAM_1135 IAM_1136 IAM_1137 IAM_1138 IAM_1139 IAM_1140 IAM_1141 IAM_1142 IAM_1143 IAM_1144 IAM_1145 IAM_1146 IAM_1147 IAM_1148 IAM_1149 IAM_1150 IAM_1151 IAM_1152 IAM_1153 IAM_1154 IAM_1155 IAM_1156 IAM_1157 IAM_1158 IAM_1159 IAM_1160 IAM_1161 IAM_1162 IAM_1163 IAM_1164 IAM_1165
network_type	2	Cognitive Non-Cognitive

Dimensions	
Covariance Parameters	28
Columns in X	4
Columns in Z	0
Subjects	157
Max Obs per Subject	7

Number of Observations	
Number of Observations Read	1099
Number of Observations Used	1099
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Res Log Like	Criterion
0	1	-324.25142608	
1	4	-2770.48086894	0.00120897
2	1	-2773.80118987	0.00010245
3	1	-2774.06516928	0.00000177
4	1	-2774.06947085	0.00000000

Convergence criteria met.

Estimated R Correlation Matrix for studyid IAM_1001							
Row	Col1	Col2	Col3	Col4	Col5	Col6	Col7
1	1.0000	0.9617	0.9597	0.8860	0.9602	0.9510	0.8594
2	0.9617	1.0000	0.9871	0.9342	0.9776	0.9101	0.8621
3	0.9597	0.9871	1.0000	0.9039	0.9707	0.8940	0.8409
4	0.8860	0.9342	0.9039	1.0000	0.9098	0.8723	0.8639
5	0.9602	0.9776	0.9707	0.9098	1.0000	0.9272	0.8427
6	0.9510	0.9101	0.8940	0.8723	0.9272	1.0000	0.8097
7	0.8594	0.8621	0.8409	0.8639	0.8427	0.8097	1.0000

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
UN(1,1)	studyid	0.04676
UN(2,1)	studyid	0.05104
UN(2,2)	studyid	0.06023

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
UN(3,1)	studyid	0.05189
UN(3,2)	studyid	0.06057
UN(3,3)	studyid	0.06251
UN(4,1)	studyid	0.03840
UN(4,2)	studyid	0.04596
UN(4,3)	studyid	0.04530
UN(4,4)	studyid	0.04018
UN(5,1)	studyid	0.04568
UN(5,2)	studyid	0.05279
UN(5,3)	studyid	0.05340
UN(5,4)	studyid	0.04012
UN(5,5)	studyid	0.04841
UN(6,1)	studyid	0.03338
UN(6,2)	studyid	0.03625
UN(6,3)	studyid	0.03628
UN(6,4)	studyid	0.02838
UN(6,5)	studyid	0.03311
UN(6,6)	studyid	0.02635
UN(7,1)	studyid	0.03733
UN(7,2)	studyid	0.04250
UN(7,3)	studyid	0.04224
UN(7,4)	studyid	0.03479
UN(7,5)	studyid	0.03725
UN(7,6)	studyid	0.02640
UN(7,7)	studyid	0.04036

Fit Statistics	
-2 Res Log Likelihood	-2774.1
AIC (Smaller is Better)	-2718.1
AICC (Smaller is Better)	-2716.5
BIC (Smaller is Better)	-2632.5

Null Model Likelihood Ratio Test		
DF	Chi-Square	Pr > ChiSq
27	2449.82	<.0001

Solution for Fixed Effects									
Effect	network_type	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
Intercept		1.0005	0.06976	155	14.34	<.0001	0.05	0.8627	1.1383
age		0.002587	0.001021	155	2.53	0.0123	0.05	0.000570	0.004604
network_type	Cognitive	0.05472	0.003680	155	14.87	<.0001	0.05	0.04745	0.06199
network_type	Non-Cognitive	0	.	.	.	.	.	.	.

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
age	1	155	6.42	0.0123
network_type	1	155	221.15	<.0001

Least Squares Means									
Effect	network_type	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
network_type	Cognitive	1.2299	0.01249	155	98.50	<.0001	0.05	1.2052	1.2546
network_type	Non-Cognitive	1.1752	0.01062	155	110.70	<.0001	0.05	1.1542	1.1961

Differences of Least Squares Means										
Effect	network_type	network_type	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
network_type	Cognitive	Non-Cognitive	0.05472	0.003680	155	14.87	<.0001	0.05	0.04745	0.06199

The SAS System

The Mixed Procedure

Model Information	
Data Set	WORK.NEW2
Dependent Variable	Amyloid
Covariance Structure	Unstructured
Subject Effect	studyid
Estimation Method	REML
Residual Variance Method	None
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Between-Within

Class Level Information		
Class	Levels	Values
studyid	157	IAM_1001 IAM_1002 IAM_1003 IAM_1004 IAM_1006 IAM_1008 IAM_1009 IAM_1010 IAM_1011 IAM_1012 IAM_1013 IAM_1014 IAM_1015 IAM_1016 IAM_1017 IAM_1018 IAM_1019 IAM_1020 IAM_1021 IAM_1022 IAM_1023 IAM_1025 IAM_1026 IAM_1027 IAM_1028 IAM_1029 IAM_1030 IAM_1031 IAM_1032 IAM_1033 IAM_1034 IAM_1035 IAM_1036 IAM_1037 IAM_1039 IAM_1040 IAM_1041 IAM_1042 IAM_1043 IAM_1044 IAM_1045 IAM_1046 IAM_1047 IAM_1048 IAM_1049 IAM_1050 IAM_1051 IAM_1052 IAM_1053 IAM_1054 IAM_1055 IAM_1056 IAM_1057 IAM_1058 IAM_1059 IAM_1062 IAM_1063 IAM_1064 IAM_1065 IAM_1066 IAM_1067 IAM_1068 IAM_1069 IAM_1071 IAM_1072 IAM_1073 IAM_1074 IAM_1075 IAM_1076 IAM_1077 IAM_1078 IAM_1079 IAM_1080 IAM_1081 IAM_1082 IAM_1083 IAM_1084 IAM_1085 IAM_1086 IAM_1087 IAM_1088 IAM_1089 IAM_1090 IAM_1091 IAM_1092 IAM_1093 IAM_1094 IAM_1095 IAM_1096 IAM_1097 IAM_1098 IAM_1099 IAM_1101 IAM_1102 IAM_1103 IAM_1104 IAM_1105 IAM_1106 IAM_1107 IAM_1108 IAM_1109 IAM_1110 IAM_1111 IAM_1112 IAM_1113 IAM_1114 IAM_1115 IAM_1116 IAM_1117 IAM_1118 IAM_1119 IAM_1120 IAM_1121 IAM_1122 IAM_1123 IAM_1124 IAM_1125 IAM_1126 IAM_1127 IAM_1128 IAM_1129 IAM_1130 IAM_1131 IAM_1132 IAM_1133 IAM_1134 IAM_1135 IAM_1136 IAM_1137 IAM_1138 IAM_1139 IAM_1140 IAM_1141 IAM_1142 IAM_1143 IAM_1144 IAM_1145 IAM_1146 IAM_1147 IAM_1148 IAM_1149 IAM_1150 IAM_1151 IAM_1152 IAM_1153 IAM_1154 IAM_1155 IAM_1156 IAM_1157 IAM_1158 IAM_1159 IAM_1160 IAM_1161 IAM_1162 IAM_1163 IAM_1164 IAM_1165
network_type	2	Cognitive Non-Cognitive

Dimensions	
Covariance Parameters	28
Columns in X	6
Columns in Z	0
Subjects	157
Max Obs per Subject	7

Number of Observations	
Number of Observations Read	1099
Number of Observations Used	1099
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Res Log Like	Criterion
0	1	-315.80146406	
1	2	-2758.04353758	0.00124034
2	1	-2761.39800863	0.00008019
3	1	-2761.60049831	0.00000079
4	1	-2761.60238903	0.00000000

Convergence criteria met.

Estimated R Correlation Matrix for studyid IAM_1001							
Row	Col1	Col2	Col3	Col4	Col5	Col6	Col7
1	1.0000	0.9606	0.9584	0.8837	0.9594	0.9510	0.8568
2	0.9606	1.0000	0.9868	0.9323	0.9770	0.9096	0.8603
3	0.9584	0.9868	1.0000	0.9013	0.9699	0.8933	0.8389
4	0.8837	0.9323	0.9013	1.0000	0.9079	0.8715	0.8614
5	0.9594	0.9770	0.9699	0.9079	1.0000	0.9270	0.8401
6	0.9510	0.9096	0.8933	0.8715	0.9270	1.0000	0.8072
7	0.8568	0.8603	0.8389	0.8614	0.8401	0.8072	1.0000

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
UN(1,1)	studyid	0.04587
UN(2,1)	studyid	0.04995
UN(2,2)	studyid	0.05895

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
UN(3,1)	studyid	0.05077
UN(3,2)	studyid	0.05926
UN(3,3)	studyid	0.06118
UN(4,1)	studyid	0.03753
UN(4,2)	studyid	0.04488
UN(4,3)	studyid	0.04420
UN(4,4)	studyid	0.03931
UN(5,1)	studyid	0.04475
UN(5,2)	studyid	0.05166
UN(5,3)	studyid	0.05224
UN(5,4)	studyid	0.03920
UN(5,5)	studyid	0.04743
UN(6,1)	studyid	0.03290
UN(6,2)	studyid	0.03567
UN(6,3)	studyid	0.03569
UN(6,4)	studyid	0.02791
UN(6,5)	studyid	0.03261
UN(6,6)	studyid	0.02609
UN(7,1)	studyid	0.03665
UN(7,2)	studyid	0.04172
UN(7,3)	studyid	0.04144
UN(7,4)	studyid	0.03411
UN(7,5)	studyid	0.03654
UN(7,6)	studyid	0.02604
UN(7,7)	studyid	0.03988

Fit Statistics	
-2 Res Log Likelihood	-2761.6
AIC (Smaller is Better)	-2705.6
AICC (Smaller is Better)	-2704.1
BIC (Smaller is Better)	-2620.0

Null Model Likelihood Ratio Test		
DF	Chi-Square	Pr > ChiSq
27	2445.80	<.0001

Solution for Fixed Effects									
Effect	network_type	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
Intercept		0.9649	0.07527	155	12.82	<.0001	0.05	0.8162	1.1135
age		0.003122	0.001103	155	2.83	0.0053	0.05	0.000943	0.005302
network_type	Cognitive	0.02220	0.02606	155	0.85	0.3954	0.05	-0.02927	0.07367
network_type	Non-Cognitive	0	.	.	.	.	.	.	.
age*network_type	Cognitive	0.000489	0.000382	155	1.28	0.2025	0.05	-0.00027	0.001243
age*network_type	Non-Cognitive	0	.	.	.	.	.	.	.

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
age	1	155	8.02	0.0052
network_type	1	155	0.73	0.3954
age*network_type	1	155	1.64	0.2025

Least Squares Means										
Effect	network_type	age	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
network_type	Cognitive	45.00	1.1496	0.03178	155	36.17	<.0001	0.05	1.0868	1.2124
network_type	Non-Cognitive	45.00	1.1054	0.02703	155	40.89	<.0001	0.05	1.0520	1.1588

Least Squares Means										
Effect	network_type	age	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
network_type	Cognitive	55.00	1.1857	0.02050	155	57.85	<.0001	0.05	1.1452	1.2262
network_type	Non-Cognitive	55.00	1.1366	0.01743	155	65.20	<.0001	0.05	1.1022	1.1710
network_type	Cognitive	65.00	1.2218	0.01290	155	94.68	<.0001	0.05	1.1963	1.2473
network_type	Non-Cognitive	65.00	1.1678	0.01098	155	106.40	<.0001	0.05	1.1461	1.1895
network_type	Cognitive	75.00	1.2579	0.01580	155	79.63	<.0001	0.05	1.2267	1.2891
network_type	Non-Cognitive	75.00	1.1990	0.01344	155	89.24	<.0001	0.05	1.1725	1.2256
network_type	Cognitive	85.00	1.2940	0.02587	155	50.02	<.0001	0.05	1.2429	1.3451
network_type	Non-Cognitive	85.00	1.2303	0.02200	155	55.92	<.0001	0.05	1.1868	1.2737

The SAS System

The Mixed Procedure

Model Information	
Data Set	WORK.NEW2
Dependent Variable	Amyloid
Covariance Structure	Unstructured
Subject Effect	studyid
Estimation Method	REML
Residual Variance Method	None
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Between-Within

Class Level Information		
Class	Levels	Values
studyid	157	IAM_1001 IAM_1002 IAM_1003 IAM_1004 IAM_1006 IAM_1008 IAM_1009 IAM_1010 IAM_1011 IAM_1012 IAM_1013 IAM_1014 IAM_1015 IAM_1016 IAM_1017 IAM_1018 IAM_1019 IAM_1020 IAM_1021 IAM_1022 IAM_1023 IAM_1025 IAM_1026 IAM_1027 IAM_1028 IAM_1029 IAM_1030 IAM_1031 IAM_1032 IAM_1033 IAM_1034 IAM_1035 IAM_1036 IAM_1037 IAM_1039 IAM_1040 IAM_1041 IAM_1042 IAM_1043 IAM_1044 IAM_1045 IAM_1046 IAM_1047 IAM_1048 IAM_1049 IAM_1050 IAM_1051 IAM_1052 IAM_1053 IAM_1054 IAM_1055 IAM_1056 IAM_1057 IAM_1058 IAM_1059 IAM_1062 IAM_1063 IAM_1064 IAM_1065 IAM_1066 IAM_1067 IAM_1068 IAM_1069 IAM_1071 IAM_1072 IAM_1073 IAM_1074 IAM_1075 IAM_1076 IAM_1077 IAM_1078 IAM_1079 IAM_1080 IAM_1081 IAM_1082 IAM_1083 IAM_1084 IAM_1085 IAM_1086 IAM_1087 IAM_1088 IAM_1089 IAM_1090 IAM_1091 IAM_1092 IAM_1093 IAM_1094 IAM_1095 IAM_1096 IAM_1097 IAM_1098 IAM_1099 IAM_1101 IAM_1102 IAM_1103 IAM_1104 IAM_1105 IAM_1106 IAM_1107 IAM_1108 IAM_1109 IAM_1110 IAM_1111 IAM_1112 IAM_1113 IAM_1114 IAM_1115 IAM_1116 IAM_1117 IAM_1118 IAM_1119 IAM_1120 IAM_1121 IAM_1122 IAM_1123 IAM_1124 IAM_1125 IAM_1126 IAM_1127 IAM_1128 IAM_1129 IAM_1130 IAM_1131 IAM_1132 IAM_1133 IAM_1134 IAM_1135 IAM_1136 IAM_1137 IAM_1138 IAM_1139 IAM_1140 IAM_1141 IAM_1142 IAM_1143 IAM_1144 IAM_1145 IAM_1146 IAM_1147 IAM_1148 IAM_1149 IAM_1150 IAM_1151 IAM_1152 IAM_1153 IAM_1154 IAM_1155 IAM_1156 IAM_1157 IAM_1158 IAM_1159 IAM_1160 IAM_1161 IAM_1162 IAM_1163 IAM_1164 IAM_1165
network_type	2	Cognitive Non-Cognitive
pet_read	2	AB+ AB-

Dimensions	
Covariance Parameters	28
Columns in X	10
Columns in Z	0
Subjects	157
Max Obs per Subject	7

Number of Observations	
Number of Observations Read	1099
Number of Observations Used	1099
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Res Log Like	Criterion
0	1	-837.22883727	
1	4	-2790.83469336	0.00195462
2	1	-2796.22722285	0.00016015
3	1	-2796.63771450	0.00000207
4	1	-2796.64274527	0.00000000

Convergence criteria met.

Estimated R Correlation Matrix for studyid IAM_1001							
Row	Col1	Col2	Col3	Col4	Col5	Col6	Col7
1	1.0000	0.9366	0.9304	0.8228	0.9345	0.9250	0.8070
2	0.9366	1.0000	0.9792	0.8813	0.9622	0.8546	0.8073
3	0.9304	0.9792	1.0000	0.8310	0.9485	0.8260	0.7749
4	0.8228	0.8813	0.8310	1.0000	0.8585	0.8146	0.8023
5	0.9345	0.9622	0.9485	0.8585	1.0000	0.8874	0.7812
6	0.9250	0.8546	0.8260	0.8146	0.8874	1.0000	0.7334
7	0.8070	0.8073	0.7749	0.8023	0.7812	0.7334	1.0000

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
UN(1,1)	studyid	0.02789
UN(2,1)	studyid	0.03050

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
UN(2,2)	studyid	0.03804
UN(3,1)	studyid	0.03064
UN(3,2)	studyid	0.03767
UN(3,3)	studyid	0.03890
UN(4,1)	studyid	0.02349
UN(4,2)	studyid	0.02938
UN(4,3)	studyid	0.02801
UN(4,4)	studyid	0.02922
UN(5,1)	studyid	0.02689
UN(5,2)	studyid	0.03234
UN(5,3)	studyid	0.03224
UN(5,4)	studyid	0.02529
UN(5,5)	studyid	0.02969
UN(6,1)	studyid	0.02067
UN(6,2)	studyid	0.02230
UN(6,3)	studyid	0.02180
UN(6,4)	studyid	0.01863
UN(6,5)	studyid	0.02046
UN(6,6)	studyid	0.01791
UN(7,1)	studyid	0.02333
UN(7,2)	studyid	0.02726
UN(7,3)	studyid	0.02646
UN(7,4)	studyid	0.02374
UN(7,5)	studyid	0.02331
UN(7,6)	studyid	0.01699
UN(7,7)	studyid	0.02998

Fit Statistics	
-2 Res Log Likelihood	-2796.6
AIC (Smaller is Better)	-2740.6
AICC (Smaller is Better)	-2739.1
BIC (Smaller is Better)	-2655.1

Null Model Likelihood Ratio Test		
DF	Chi-Square	Pr > ChiSq
27	1959.41	<.0001

Solution for Fixed Effects										
Effect	network_type	pet_read	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
Intercept			0.9705	0.06774	154	14.33	<.0001	0.05	0.8367	1.1043
age			0.002657	0.001003	154	2.65	0.0089	0.05	0.000674	0.004639
pet_read		AB+	0.1790	0.02620	154	6.83	<.0001	0.05	0.1273	0.2308
pet_read		AB-	0	.	.	.	.	.	.	.
network_type	Cognitive		0.04615	0.003782	154	12.20	<.0001	0.05	0.03868	0.05362
network_type	Non-Cognitive		0	.	.	.	.	.	.	.
network_typ*pet_read	Cognitive	AB+	0.06052	0.008956	154	6.76	<.0001	0.05	0.04283	0.07821
network_typ*pet_read	Cognitive	AB-	0	.	.	.	.	.	.	.
network_typ*pet_read	Non-Cognitive	AB+	0	.	.	.	.	.	.	.
network_typ*pet_read	Non-Cognitive	AB-	0	.	.	.	.	.	.	.

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
age	1	154	7.01	0.0089
pet_read	1	154	56.84	<.0001
network_type	1	154	291.17	<.0001

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
network_typ*pet_read	1	154	45.67	<.0001

Least Squares Means										
Effect	network_type	pet_read	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
network_typ*pet_read	Cognitive	AB+	1.4356	0.02704	154	53.08	<.0001	0.05	1.3822	1.4890
network_typ*pet_read	Cognitive	AB-	1.1960	0.01249	154	95.79	<.0001	0.05	1.1714	1.2207
network_typ*pet_read	Non-Cognitive	AB+	1.3289	0.02368	154	56.13	<.0001	0.05	1.2821	1.3757
network_typ*pet_read	Non-Cognitive	AB-	1.1499	0.01090	154	105.50	<.0001	0.05	1.1284	1.1714

Differences of Least Squares Means																
Effect	network_type	pet_read	network_type	pet_read	Estimate	Standard Error	DF	t Value	Pr > t	Adjustment	Adj P	Alpha	Lower	Upper	Adj Lower	Adj Upper
network_typ*pet_read	Cognitive	AB+	Cognitive	AB-	0.2395	0.02991	154	8.01	<.0001	Tukey-Kramer	<.0001	0.05	0.1804	0.2986	0.1618	0.3172
network_typ*pet_read	Cognitive	AB+	Non-Cognitive	AB+	0.1067	0.008118	154	13.14	<.0001	Tukey-Kramer	<.0001	0.05	0.09063	0.1227	0.08558	0.1278
network_typ*pet_read	Cognitive	AB+	Non-Cognitive	AB-	0.2857	0.02928	154	9.76	<.0001	Tukey-Kramer	<.0001	0.05	0.2278	0.3435	0.2096	0.3617
network_typ*pet_read	Cognitive	AB-	Non-Cognitive	AB+	-0.1329	0.02690	154	-4.94	<.0001	Tukey-Kramer	<.0001	0.05	-0.1860	-0.07972	-0.2027	-0.06300
network_typ*pet_read	Cognitive	AB-	Non-Cognitive	AB-	0.04615	0.003782	154	12.20	<.0001	Tukey-Kramer	<.0001	0.05	0.03868	0.05362	0.03632	0.05597
network_typ*pet_read	Non-Cognitive	AB+	Non-Cognitive	AB-	0.1790	0.02620	154	6.83	<.0001	Tukey-Kramer	<.0001	0.05	0.1273	0.2308	0.1110	0.2471

The SAS System

The Mixed Procedure

Model Information	
Data Set	WORK.NEW2
Dependent Variable	Amyloid
Covariance Structure	Unstructured
Subject Effect	studyid
Estimation Method	REML
Residual Variance Method	None
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Between-Within

Class Level Information		
Class	Levels	Values
studyid	157	IAM_1001 IAM_1002 IAM_1003 IAM_1004 IAM_1006 IAM_1008 IAM_1009 IAM_1010 IAM_1011 IAM_1012 IAM_1013 IAM_1014 IAM_1015 IAM_1016 IAM_1017 IAM_1018 IAM_1019 IAM_1020 IAM_1021 IAM_1022 IAM_1023 IAM_1025 IAM_1026 IAM_1027 IAM_1028 IAM_1029 IAM_1030 IAM_1031 IAM_1032 IAM_1033 IAM_1034 IAM_1035 IAM_1036 IAM_1037 IAM_1039 IAM_1040 IAM_1041 IAM_1042 IAM_1043 IAM_1044 IAM_1045 IAM_1046 IAM_1047 IAM_1048 IAM_1049 IAM_1050 IAM_1051 IAM_1052 IAM_1053 IAM_1054 IAM_1055 IAM_1056 IAM_1057 IAM_1058 IAM_1059 IAM_1062 IAM_1063 IAM_1064 IAM_1065 IAM_1066 IAM_1067 IAM_1068 IAM_1069 IAM_1071 IAM_1072 IAM_1073 IAM_1074 IAM_1075 IAM_1076 IAM_1077 IAM_1078 IAM_1079 IAM_1080 IAM_1081 IAM_1082 IAM_1083 IAM_1084 IAM_1085 IAM_1086 IAM_1087 IAM_1088 IAM_1089 IAM_1090 IAM_1091 IAM_1092 IAM_1093 IAM_1094 IAM_1095 IAM_1096 IAM_1097 IAM_1098 IAM_1099 IAM_1101 IAM_1102 IAM_1103 IAM_1104 IAM_1105 IAM_1106 IAM_1107 IAM_1108 IAM_1109 IAM_1110 IAM_1111 IAM_1112 IAM_1113 IAM_1114 IAM_1115 IAM_1116 IAM_1117 IAM_1118 IAM_1119 IAM_1120 IAM_1121 IAM_1122 IAM_1123 IAM_1124 IAM_1125 IAM_1126 IAM_1127 IAM_1128 IAM_1129 IAM_1130 IAM_1131 IAM_1132 IAM_1133 IAM_1134 IAM_1135 IAM_1136 IAM_1137 IAM_1138 IAM_1139 IAM_1140 IAM_1141 IAM_1142 IAM_1143 IAM_1144 IAM_1145 IAM_1146 IAM_1147 IAM_1148 IAM_1149 IAM_1150 IAM_1151 IAM_1152 IAM_1153 IAM_1154 IAM_1155 IAM_1156 IAM_1157 IAM_1158 IAM_1159 IAM_1160 IAM_1161 IAM_1162 IAM_1163 IAM_1164 IAM_1165
Network	7	DAT DMN FPN LIM SOM VAT VIS

Dimensions	
Covariance Parameters	28
Columns in X	9
Columns in Z	0
Subjects	157
Max Obs per Subject	7

Number of Observations	
Number of Observations Read	1099
Number of Observations Used	1099
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Res Log Like	Criterion
0	1	-306.60169765	
1	2	-2883.52023167	0.00101560
2	1	-2886.40054073	0.00009159
3	1	-2886.63893844	0.00000100
4	1	-2886.64140831	0.00000000

Convergence criteria met.

Estimated R Correlation Matrix for studyid IAM_1001							
Row	Col1	Col2	Col3	Col4	Col5	Col6	Col7
1	1.0000	0.9635	0.9620	0.8865	0.9654	0.9550	0.8971
2	0.9635	1.0000	0.9869	0.9330	0.9779	0.9218	0.8803
3	0.9620	0.9869	1.0000	0.9020	0.9705	0.9068	0.8551
4	0.8865	0.9330	0.9020	1.0000	0.9084	0.8830	0.8829
5	0.9654	0.9779	0.9705	0.9084	1.0000	0.9459	0.8490
6	0.9550	0.9218	0.9068	0.8830	0.9459	1.0000	0.8766
7	0.8971	0.8803	0.8551	0.8829	0.8490	0.8766	1.0000

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
UN(1,1)	studyid	0.04672
UN(2,1)	studyid	0.05078
UN(2,2)	studyid	0.05946

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
UN(3,1)	studyid	0.05160
UN(3,2)	studyid	0.05971
UN(3,3)	studyid	0.06157
UN(4,1)	studyid	0.03816
UN(4,2)	studyid	0.04530
UN(4,3)	studyid	0.04457
UN(4,4)	studyid	0.03964
UN(5,1)	studyid	0.04532
UN(5,2)	studyid	0.05179
UN(5,3)	studyid	0.05230
UN(5,4)	studyid	0.03928
UN(5,5)	studyid	0.04717
UN(6,1)	studyid	0.03353
UN(6,2)	studyid	0.03651
UN(6,3)	studyid	0.03655
UN(6,4)	studyid	0.02855
UN(6,5)	studyid	0.03337
UN(6,6)	studyid	0.02638
UN(7,1)	studyid	0.03649
UN(7,2)	studyid	0.04039
UN(7,3)	studyid	0.03993
UN(7,4)	studyid	0.03308
UN(7,5)	studyid	0.03470
UN(7,6)	studyid	0.02679
UN(7,7)	studyid	0.03541

Fit Statistics	
-2 Res Log Likelihood	-2886.6
AIC (Smaller is Better)	-2830.6
AICC (Smaller is Better)	-2829.1
BIC (Smaller is Better)	-2745.1

Null Model Likelihood Ratio Test		
DF	Chi-Square	Pr > ChiSq
27	2580.04	<.0001

Solution for Fixed Effects									
Effect	Network	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
Intercept		1.0716	0.07057	155	15.19	<.0001	0.05	0.9322	1.2110
age		0.002587	0.001021	155	2.53	0.0123	0.05	0.000570	0.004604
Network	DAT	-0.00304	0.007637	155	-0.40	0.6908	0.05	-0.01813	0.01204
Network	DMN	0.01513	0.009473	155	1.60	0.1122	0.05	-0.00358	0.03385
Network	FPN	0.01788	0.01045	155	1.71	0.0889	0.05	-0.00275	0.03852
Network	LIM	0.008773	0.007525	155	1.17	0.2455	0.05	-0.00609	0.02364
Network	SOM	-0.07577	0.007231	155	-10.48	<.0001	0.05	-0.09006	-0.06149
Network	VAT	0.02077	0.009165	155	2.27	0.0248	0.05	0.002667	0.03888
Network	VIS	0	.	.	.	.	.	.	.

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
age	1	155	6.42	0.0123
Network	6	155	81.53	<.0001

Least Squares Means									
Effect	Network	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
Network	DAT	1.2432	0.01725	155	72.07	<.0001	0.05	1.2092	1.2773
Network	DMN	1.2614	0.01946	155	64.82	<.0001	0.05	1.2230	1.2999
Network	FPN	1.2642	0.01980	155	63.83	<.0001	0.05	1.2250	1.3033
Network	LIM	1.2551	0.01589	155	78.98	<.0001	0.05	1.2237	1.2864
Network	SOM	1.1705	0.01296	155	90.30	<.0001	0.05	1.1449	1.1961
Network	VAT	1.2671	0.01733	155	73.10	<.0001	0.05	1.2328	1.3013
Network	VIS	1.2463	0.01502	155	82.98	<.0001	0.05	1.2166	1.2760

Differences of Least Squares Means														
Effect	Network	Network	Estimate	Standard Error	DF	t Value	Pr > t	Adjustment	Adj P	Alpha	Lower	Upper	Adj Lower	Adj Upper
Network	DAT	DMN	-0.01818	0.005423	155	-3.35	0.0010	Tukey-Kramer	0.0171	0.05	-0.02889	-0.00746	-0.03438	-0.00197
Network	DAT	FPN	-0.02092	0.005698	155	-3.67	0.0003	Tukey-Kramer	0.0060	0.05	-0.03218	-0.00967	-0.03795	-0.00390
Network	DAT	LIM	-0.01182	0.008003	155	-1.48	0.1418	Tukey-Kramer	0.7583	0.05	-0.02763	0.003993	-0.03573	0.01209
Network	DAT	SOM	0.07273	0.006208	155	11.72	<.0001	Tukey-Kramer	<.0001	0.05	0.06047	0.08499	0.05418	0.09128
Network	DAT	VAT	-0.02381	0.004552	155	-5.23	<.0001	Tukey-Kramer	<.0001	0.05	-0.03281	-0.01482	-0.03741	-0.01021
Network	DAT	VIS	-0.00304	0.007637	155	-0.40	0.6908	Tukey-Kramer	0.9997	0.05	-0.01813	0.01204	-0.02586	0.01977
Network	DMN	FPN	-0.00275	0.003201	155	-0.86	0.3918	Tukey-Kramer	0.9780	0.05	-0.00907	0.003575	-0.01231	0.006815
Network	DMN	LIM	0.006359	0.007363	155	0.86	0.3891	Tukey-Kramer	0.9773	0.05	-0.00819	0.02090	-0.01564	0.02836
Network	DMN	SOM	0.09091	0.009039	155	10.06	<.0001	Tukey-Kramer	<.0001	0.05	0.07305	0.1088	0.06390	0.1179
Network	DMN	VAT	-0.00564	0.004405	155	-1.28	0.2025	Tukey-Kramer	0.8602	0.05	-0.01434	0.003064	-0.01880	0.007523
Network	DMN	VIS	0.01513	0.009473	155	1.60	0.1122	Tukey-Kramer	0.6841	0.05	-0.00358	0.03385	-0.01317	0.04343
Network	FPN	LIM	0.009108	0.008774	155	1.04	0.3008	Tukey-Kramer	0.9443	0.05	-0.00822	0.02644	-0.01710	0.03532
Network	FPN	SOM	0.09365	0.009728	155	9.63	<.0001	Tukey-Kramer	<.0001	0.05	0.07444	0.1129	0.06459	0.1227
Network	FPN	VAT	-0.00289	0.005135	155	-0.56	0.5744	Tukey-Kramer	0.9977	0.05	-0.01303	0.007253	-0.01823	0.01245
Network	FPN	VIS	0.01788	0.01045	155	1.71	0.0889	Tukey-Kramer	0.6092	0.05	-0.00275	0.03852	-0.01333	0.04909
Network	LIM	SOM	0.08455	0.007535	155	11.22	<.0001	Tukey-Kramer	<.0001	0.05	0.06966	0.09943	0.06203	0.1071
Network	LIM	VAT	-0.01200	0.007248	155	-1.66	0.0999	Tukey-Kramer	0.6466	0.05	-0.02631	0.002319	-0.03365	0.009655
Network	LIM	VIS	0.008773	0.007525	155	1.17	0.2455	Tukey-Kramer	0.9058	0.05	-0.00609	0.02364	-0.01371	0.03126
Network	SOM	VAT	-0.09654	0.006588	155	-14.66	<.0001	Tukey-Kramer	<.0001	0.05	-0.1096	-0.08353	-0.1162	-0.07686
Network	SOM	VIS	-0.07577	0.007231	155	-10.48	<.0001	Tukey-Kramer	<.0001	0.05	-0.09006	-0.06149	-0.09737	-0.05417
Network	VAT	VIS	0.02077	0.009165	155	2.27	0.0248	Tukey-Kramer	0.2672	0.05	0.002667	0.03888	-0.00661	0.04815

The SAS System

The Mixed Procedure

Model Information	
Data Set	WORK.NEW2
Dependent Variable	Amyloid
Covariance Structure	Unstructured
Subject Effect	studyid
Estimation Method	REML
Residual Variance Method	None
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Between-Within

Class Level Information		
Class	Levels	Values
studyid	157	IAM_1001 IAM_1002 IAM_1003 IAM_1004 IAM_1006 IAM_1008 IAM_1009 IAM_1010 IAM_1011 IAM_1012 IAM_1013 IAM_1014 IAM_1015 IAM_1016 IAM_1017 IAM_1018 IAM_1019 IAM_1020 IAM_1021 IAM_1022 IAM_1023 IAM_1025 IAM_1026 IAM_1027 IAM_1028 IAM_1029 IAM_1030 IAM_1031 IAM_1032 IAM_1033 IAM_1034 IAM_1035 IAM_1036 IAM_1037 IAM_1039 IAM_1040 IAM_1041 IAM_1042 IAM_1043 IAM_1044 IAM_1045 IAM_1046 IAM_1047 IAM_1048 IAM_1049 IAM_1050 IAM_1051 IAM_1052 IAM_1053 IAM_1054 IAM_1055 IAM_1056 IAM_1057 IAM_1058 IAM_1059 IAM_1062 IAM_1063 IAM_1064 IAM_1065 IAM_1066 IAM_1067 IAM_1068 IAM_1069 IAM_1071 IAM_1072 IAM_1073 IAM_1074 IAM_1075 IAM_1076 IAM_1077 IAM_1078 IAM_1079 IAM_1080 IAM_1081 IAM_1082 IAM_1083 IAM_1084 IAM_1085 IAM_1086 IAM_1087 IAM_1088 IAM_1089 IAM_1090 IAM_1091 IAM_1092 IAM_1093 IAM_1094 IAM_1095 IAM_1096 IAM_1097 IAM_1098 IAM_1099 IAM_1101 IAM_1102 IAM_1103 IAM_1104 IAM_1105 IAM_1106 IAM_1107 IAM_1108 IAM_1109 IAM_1110 IAM_1111 IAM_1112 IAM_1113 IAM_1114 IAM_1115 IAM_1116 IAM_1117 IAM_1118 IAM_1119 IAM_1120 IAM_1121 IAM_1122 IAM_1123 IAM_1124 IAM_1125 IAM_1126 IAM_1127 IAM_1128 IAM_1129 IAM_1130 IAM_1131 IAM_1132 IAM_1133 IAM_1134 IAM_1135 IAM_1136 IAM_1137 IAM_1138 IAM_1139 IAM_1140 IAM_1141 IAM_1142 IAM_1143 IAM_1144 IAM_1145 IAM_1146 IAM_1147 IAM_1148 IAM_1149 IAM_1150 IAM_1151 IAM_1152 IAM_1153 IAM_1154 IAM_1155 IAM_1156 IAM_1157 IAM_1158 IAM_1159 IAM_1160 IAM_1161 IAM_1162 IAM_1163 IAM_1164 IAM_1165
Network	7	DAT DMN FPN LIM SOM VAT VIS

Dimensions	
Covariance Parameters	28
Columns in X	16
Columns in Z	0
Subjects	157
Max Obs per Subject	7

Number of Observations	
Number of Observations Read	1099
Number of Observations Used	1099
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Res Log Like	Criterion
0	1	-247.91773526	
1	1	-2827.55288120	0.00000000

Convergence criteria met.

Estimated R Correlation Matrix for studyid IAM_1001							
Row	Col1	Col2	Col3	Col4	Col5	Col6	Col7
1	1.0000	0.9619	0.9606	0.8791	0.9631	0.9545	0.8910
2	0.9619	1.0000	0.9856	0.9284	0.9771	0.9228	0.8735
3	0.9606	0.9856	1.0000	0.8951	0.9692	0.9073	0.8465
4	0.8791	0.9284	0.8951	1.0000	0.9022	0.8791	0.8758
5	0.9631	0.9771	0.9692	0.9022	1.0000	0.9454	0.8398
6	0.9545	0.9228	0.9073	0.8791	0.9454	1.0000	0.8717
7	0.8910	0.8735	0.8465	0.8758	0.8398	0.8717	1.0000

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
UN(1,1)	studyid	0.04422
UN(2,1)	studyid	0.04737
UN(2,2)	studyid	0.05483
UN(3,1)	studyid	0.04807
UN(3,2)	studyid	0.05492
UN(3,3)	studyid	0.05662

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
UN(4,1)	studyid	0.03572
UN(4,2)	studyid	0.04200
UN(4,3)	studyid	0.04115
UN(4,4)	studyid	0.03733
UN(5,1)	studyid	0.04271
UN(5,2)	studyid	0.04825
UN(5,3)	studyid	0.04864
UN(5,4)	studyid	0.03676
UN(5,5)	studyid	0.04447
UN(6,1)	studyid	0.03208
UN(6,2)	studyid	0.03454
UN(6,3)	studyid	0.03451
UN(6,4)	studyid	0.02715
UN(6,5)	studyid	0.03187
UN(6,6)	studyid	0.02555
UN(7,1)	studyid	0.03440
UN(7,2)	studyid	0.03755
UN(7,3)	studyid	0.03698
UN(7,4)	studyid	0.03107
UN(7,5)	studyid	0.03251
UN(7,6)	studyid	0.02558
UN(7,7)	studyid	0.03371

Fit Statistics	
-2 Res Log Likelihood	-2827.6
AIC (Smaller is Better)	-2771.6
AICC (Smaller is Better)	-2770.0
BIC (Smaller is Better)	-2686.0

Null Model Likelihood Ratio Test		
DF	Chi-Square	Pr > ChiSq
27	2579.64	<.0001

Solution for Fixed Effects									
Effect	Network	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
Intercept		0.7729	0.1039	155	7.44	<.0001	0.05	0.5676	0.9781
age		0.007010	0.001523	155	4.60	<.0001	0.05	0.004001	0.01002
Network	DAT	-0.06720	0.05408	155	-1.24	0.2159	0.05	-0.1740	0.03963
Network	DMN	-0.1752	0.06560	155	-2.67	0.0084	0.05	-0.3048	-0.04563
Network	FPN	-0.1888	0.07240	155	-2.61	0.0100	0.05	-0.3318	-0.04577
Network	LIM	-0.03961	0.05339	155	-0.74	0.4593	0.05	-0.1451	0.06586
Network	SOM	0.01367	0.05093	155	0.27	0.7887	0.05	-0.08693	0.1143
Network	VAT	-0.05666	0.06490	155	-0.87	0.3840	0.05	-0.1849	0.07154
Network	VIS	0	.	.	.	.	.	.	.
age*Network	DAT	0.000950	0.000793	155	1.20	0.2326	0.05	-0.00062	0.002516
age*Network	DMN	0.002818	0.000962	155	2.93	0.0039	0.05	0.000919	0.004718
age*Network	FPN	0.003060	0.001061	155	2.88	0.0045	0.05	0.000964	0.005157
age*Network	LIM	0.000716	0.000783	155	0.92	0.3614	0.05	-0.00083	0.002263
age*Network	SOM	-0.00132	0.000747	155	-1.77	0.0780	0.05	-0.00280	0.000150
age*Network	VAT	0.001147	0.000951	155	1.21	0.2300	0.05	-0.00073	0.003026
age*Network	VIS	0	.	.	.	.	.	.	.

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
age	1	155	24.30	<.0001

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
Network	6	155	3.70	0.0018
age*Network	6	155	3.91	0.0012

Least Squares Means										
Effect	Network	age	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
Network	DAT	45.00	1.0639	0.04274	155	24.89	<.0001	0.05	0.9794	1.1483
Network	DMN	45.00	1.0400	0.04760	155	21.85	<.0001	0.05	0.9459	1.1340
Network	FPN	45.00	1.0373	0.04837	155	21.44	<.0001	0.05	0.9417	1.1328
Network	LIM	45.00	1.0810	0.03927	155	27.53	<.0001	0.05	1.0034	1.1585
Network	SOM	45.00	1.0424	0.03249	155	32.08	<.0001	0.05	0.9782	1.1066
Network	VAT	45.00	1.0833	0.04286	155	25.27	<.0001	0.05	0.9986	1.1679
Network	VIS	45.00	1.0883	0.03732	155	29.16	<.0001	0.05	1.0146	1.1621
Network	DAT	55.00	1.1435	0.02756	155	41.48	<.0001	0.05	1.0890	1.1979
Network	DMN	55.00	1.1382	0.03069	155	37.08	<.0001	0.05	1.0776	1.1989
Network	FPN	55.00	1.1380	0.03119	155	36.48	<.0001	0.05	1.0763	1.1996
Network	LIM	55.00	1.1582	0.02532	155	45.74	<.0001	0.05	1.1082	1.2082
Network	SOM	55.00	1.0993	0.02095	155	52.46	<.0001	0.05	1.0579	1.1406
Network	VAT	55.00	1.1648	0.02764	155	42.14	<.0001	0.05	1.1102	1.2194
Network	VIS	55.00	1.1584	0.02407	155	48.14	<.0001	0.05	1.1109	1.2060
Network	DAT	65.00	1.2231	0.01736	155	70.47	<.0001	0.05	1.1888	1.2574
Network	DMN	65.00	1.2365	0.01932	155	63.99	<.0001	0.05	1.1983	1.2747
Network	FPN	65.00	1.2387	0.01964	155	63.07	<.0001	0.05	1.1999	1.2775
Network	LIM	65.00	1.2355	0.01594	155	77.49	<.0001	0.05	1.2040	1.2670
Network	SOM	65.00	1.1561	0.01319	155	87.63	<.0001	0.05	1.1301	1.1822
Network	VAT	65.00	1.2464	0.01740	155	71.62	<.0001	0.05	1.2120	1.2808
Network	VIS	65.00	1.2285	0.01515	155	81.08	<.0001	0.05	1.1986	1.2585
Network	DAT	75.00	1.3027	0.02125	155	61.31	<.0001	0.05	1.2607	1.3446
Network	DMN	75.00	1.3348	0.02366	155	56.42	<.0001	0.05	1.2881	1.3815
Network	FPN	75.00	1.3394	0.02404	155	55.71	<.0001	0.05	1.2919	1.3869
Network	LIM	75.00	1.3128	0.01952	155	67.26	<.0001	0.05	1.2742	1.3513
Network	SOM	75.00	1.2130	0.01615	155	75.11	<.0001	0.05	1.1811	1.2449
Network	VAT	75.00	1.3280	0.02130	155	62.33	<.0001	0.05	1.2859	1.3700
Network	VIS	75.00	1.2986	0.01855	155	70.01	<.0001	0.05	1.2620	1.3353
Network	DAT	85.00	1.3823	0.03479	155	39.73	<.0001	0.05	1.3136	1.4510
Network	DMN	85.00	1.4331	0.03874	155	36.99	<.0001	0.05	1.3566	1.5096
Network	FPN	85.00	1.4401	0.03937	155	36.58	<.0001	0.05	1.3623	1.5178
Network	LIM	85.00	1.3900	0.03196	155	43.49	<.0001	0.05	1.3269	1.4532
Network	SOM	85.00	1.2698	0.02644	155	48.02	<.0001	0.05	1.2176	1.3221
Network	VAT	85.00	1.4095	0.03489	155	40.40	<.0001	0.05	1.3406	1.4784
Network	VIS	85.00	1.3687	0.03037	155	45.06	<.0001	0.05	1.3087	1.4287
Network	DAT	67.53	1.2432	0.01678	155	74.08	<.0001	0.05	1.2101	1.2764
Network	DMN	67.53	1.2614	0.01869	155	67.50	<.0001	0.05	1.2245	1.2983
Network	FPN	67.53	1.2642	0.01899	155	66.57	<.0001	0.05	1.2267	1.3017
Network	LIM	67.53	1.2551	0.01542	155	81.40	<.0001	0.05	1.2246	1.2855
Network	SOM	67.53	1.1705	0.01276	155	91.75	<.0001	0.05	1.1453	1.1957
Network	VAT	67.53	1.2671	0.01683	155	75.29	<.0001	0.05	1.2338	1.3003
Network	VIS	67.53	1.2463	0.01465	155	85.05	<.0001	0.05	1.2173	1.2752

Differences of Least Squares Means															
Effect	Network	Network	age	Estimate	Standard Error	DF	t Value	Pr > t	Adjustment	Adj P	Alpha	Lower	Upper	Adj Lower	Adj Upper
Network	DAT	DMN	67.53	-0.01818	0.005246	155	-3.47	0.0007	Tukey-Kramer	0.0119	0.05	-0.02854	-0.00781	-0.03385	-0.00251
Network	DAT	FPN	67.53	-0.02092	0.005479	155	-3.82	0.0002	Tukey-Kramer	0.0036	0.05	-0.03175	-0.01010	-0.03729	-0.00456
Network	DAT	LIM	67.53	-0.01182	0.008027	155	-1.47	0.1430	Tukey-Kramer	0.7609	0.05	-0.02767	0.004040	-0.03580	0.01216
Network	DAT	SOM	67.53	0.07273	0.005975	155	12.17	<.0001	Tukey-Kramer	<.0001	0.05	0.06093	0.08453	0.05488	0.09058

Differences of Least Squares Means															
Effect	Network	Network	age	Estimate	Standard Error	DF	t Value	Pr > t	Adjustment	Adj P	Alpha	Lower	Upper	Adj Lower	Adj Upper
Network	DAT	VAT	67.53	-0.02381	0.004564	155	-5.22	<.0001	Tukey-Kramer	<.0001	0.05	-0.03283	-0.01480	-0.03745	-0.01018
Network	DAT	VIS	67.53	-0.00304	0.007627	155	-0.40	0.6904	Tukey-Kramer	0.9997	0.05	-0.01811	0.01202	-0.02583	0.01974
Network	DMN	FPN	67.53	-0.00275	0.003206	155	-0.86	0.3925	Tukey-Kramer	0.9781	0.05	-0.00908	0.003584	-0.01233	0.006829
Network	DMN	LIM	67.53	0.006359	0.007206	155	0.88	0.3789	Tukey-Kramer	0.9747	0.05	-0.00788	0.02059	-0.01517	0.02789
Network	DMN	SOM	67.53	0.09091	0.008485	155	10.71	<.0001	Tukey-Kramer	<.0001	0.05	0.07414	0.1077	0.06556	0.1163
Network	DMN	VAT	67.53	-0.00564	0.004227	155	-1.33	0.1842	Tukey-Kramer	0.8349	0.05	-0.01399	0.002711	-0.01827	0.006988
Network	DMN	VIS	67.53	0.01513	0.009250	155	1.64	0.1039	Tukey-Kramer	0.6594	0.05	-0.00314	0.03341	-0.01250	0.04277
Network	FPN	LIM	67.53	0.009108	0.008614	155	1.06	0.2920	Tukey-Kramer	0.9393	0.05	-0.00791	0.02612	-0.01662	0.03484
Network	FPN	SOM	67.53	0.09365	0.009153	155	10.23	<.0001	Tukey-Kramer	<.0001	0.05	0.07557	0.1117	0.06631	0.1210
Network	FPN	VAT	67.53	-0.00289	0.004934	155	-0.59	0.5590	Tukey-Kramer	0.9971	0.05	-0.01264	0.006858	-0.01763	0.01185
Network	FPN	VIS	67.53	0.01788	0.01021	155	1.75	0.0818	Tukey-Kramer	0.5827	0.05	-0.00229	0.03805	-0.01262	0.04838
Network	LIM	SOM	67.53	0.08455	0.007393	155	11.44	<.0001	Tukey-Kramer	<.0001	0.05	0.06994	0.09915	0.06246	0.1066
Network	LIM	VAT	67.53	-0.01200	0.007263	155	-1.65	0.1006	Tukey-Kramer	0.6490	0.05	-0.02635	0.002351	-0.03370	0.009702
Network	LIM	VIS	67.53	0.008773	0.007529	155	1.17	0.2457	Tukey-Kramer	0.9060	0.05	-0.00610	0.02365	-0.01372	0.03127
Network	SOM	VAT	67.53	-0.09654	0.006327	155	-15.26	<.0001	Tukey-Kramer	<.0001	0.05	-0.1090	-0.08405	-0.1154	-0.07764
Network	SOM	VIS	67.53	-0.07577	0.007182	155	-10.55	<.0001	Tukey-Kramer	<.0001	0.05	-0.08996	-0.06159	-0.09723	-0.05432
Network	VAT	VIS	67.53	0.02077	0.009152	155	2.27	0.0246	Tukey-Kramer	0.2656	0.05	0.002693	0.03885	-0.00657	0.04811

The SAS System

The Mixed Procedure

Model Information	
Data Set	WORK.NEW2
Dependent Variable	Amyloid
Covariance Structure	Unstructured
Subject Effect	studyid
Estimation Method	REML
Residual Variance Method	None
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Between-Within

Class Level Information		
Class	Levels	Values
studyid	157	IAM_1001 IAM_1002 IAM_1003 IAM_1004 IAM_1006 IAM_1008 IAM_1009 IAM_1010 IAM_1011 IAM_1012 IAM_1013 IAM_1014 IAM_1015 IAM_1016 IAM_1017 IAM_1018 IAM_1019 IAM_1020 IAM_1021 IAM_1022 IAM_1023 IAM_1025 IAM_1026 IAM_1027 IAM_1028 IAM_1029 IAM_1030 IAM_1031 IAM_1032 IAM_1033 IAM_1034 IAM_1035 IAM_1036 IAM_1037 IAM_1039 IAM_1040 IAM_1041 IAM_1042 IAM_1043 IAM_1044 IAM_1045 IAM_1046 IAM_1047 IAM_1048 IAM_1049 IAM_1050 IAM_1051 IAM_1052 IAM_1053 IAM_1054 IAM_1055 IAM_1056 IAM_1057 IAM_1058 IAM_1059 IAM_1062 IAM_1063 IAM_1064 IAM_1065 IAM_1066 IAM_1067 IAM_1068 IAM_1069 IAM_1071 IAM_1072 IAM_1073 IAM_1074 IAM_1075 IAM_1076 IAM_1077 IAM_1078 IAM_1079 IAM_1080 IAM_1081 IAM_1082 IAM_1083 IAM_1084 IAM_1085 IAM_1086 IAM_1087 IAM_1088 IAM_1089 IAM_1090 IAM_1091 IAM_1092 IAM_1093 IAM_1094 IAM_1095 IAM_1096 IAM_1097 IAM_1098 IAM_1099 IAM_1101 IAM_1102 IAM_1103 IAM_1104 IAM_1105 IAM_1106 IAM_1107 IAM_1108 IAM_1109 IAM_1110 IAM_1111 IAM_1112 IAM_1113 IAM_1114 IAM_1115 IAM_1116 IAM_1117 IAM_1118 IAM_1119 IAM_1120 IAM_1121 IAM_1122 IAM_1123 IAM_1124 IAM_1125 IAM_1126 IAM_1127 IAM_1128 IAM_1129 IAM_1130 IAM_1131 IAM_1132 IAM_1133 IAM_1134 IAM_1135 IAM_1136 IAM_1137 IAM_1138 IAM_1139 IAM_1140 IAM_1141 IAM_1142 IAM_1143 IAM_1144 IAM_1145 IAM_1146 IAM_1147 IAM_1148 IAM_1149 IAM_1150 IAM_1151 IAM_1152 IAM_1153 IAM_1154 IAM_1155 IAM_1156 IAM_1157 IAM_1158 IAM_1159 IAM_1160 IAM_1161 IAM_1162 IAM_1163 IAM_1164 IAM_1165
Network	7	DAT DMN FPN LIM SOM VAT VIS
pet_read	2	AB+ AB-

Dimensions	
Covariance Parameters	28
Columns in X	25
Columns in Z	0
Subjects	157
Max Obs per Subject	7

Number of Observations	
Number of Observations Read	1099
Number of Observations Used	1099
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Res Log Like	Criterion
0	1	-819.04478010	
1	2	-2969.61566695	0.00000988
2	1	-2969.64070894	0.00000001
3	1	-2969.64073771	0.00000000

Convergence criteria met.

Estimated R Correlation Matrix for studyid IAM_1001							
Row	Col1	Col2	Col3	Col4	Col5	Col6	Col7
1	1.0000	0.9318	0.9272	0.8510	0.9362	0.9339	0.8542
2	0.9318	1.0000	0.9758	0.9222	0.9597	0.8755	0.8247
3	0.9272	0.9758	1.0000	0.8782	0.9458	0.8515	0.7855
4	0.8510	0.9222	0.8782	1.0000	0.8814	0.8343	0.8357
5	0.9362	0.9597	0.9458	0.8814	1.0000	0.9159	0.7733
6	0.9339	0.8755	0.8515	0.8343	0.9159	1.0000	0.8171
7	0.8542	0.8247	0.7855	0.8357	0.7733	0.8171	1.0000

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
UN(1,1)	studyid	0.02475
UN(2,1)	studyid	0.02640
UN(2,2)	studyid	0.03242

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
UN(3,1)	studyid	0.02610
UN(3,2)	studyid	0.03144
UN(3,3)	studyid	0.03202
UN(4,1)	studyid	0.02277
UN(4,2)	studyid	0.02824
UN(4,3)	studyid	0.02672
UN(4,4)	studyid	0.02892
UN(5,1)	studyid	0.02382
UN(5,2)	studyid	0.02794
UN(5,3)	studyid	0.02737
UN(5,4)	studyid	0.02424
UN(5,5)	studyid	0.02615
UN(6,1)	studyid	0.01916
UN(6,2)	studyid	0.02055
UN(6,3)	studyid	0.01986
UN(6,4)	studyid	0.01850
UN(6,5)	studyid	0.01931
UN(6,6)	studyid	0.01700
UN(7,1)	studyid	0.02113
UN(7,2)	studyid	0.02334
UN(7,3)	studyid	0.02210
UN(7,4)	studyid	0.02234
UN(7,5)	studyid	0.01966
UN(7,6)	studyid	0.01675
UN(7,7)	studyid	0.02472

Fit Statistics	
-2 Res Log Likelihood	-2969.6
AIC (Smaller is Better)	-2913.6
AICC (Smaller is Better)	-2912.1
BIC (Smaller is Better)	-2828.1

Null Model Likelihood Ratio Test		
DF	Chi-Square	Pr > ChiSq
27	2150.60	<.0001

Solution for Fixed Effects										
Effect	Network	pet_read	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
Intercept			1.0186	0.06829	154	14.92	<.0001	0.05	0.8837	1.1535
age			0.002656	0.001003	154	2.65	0.0089	0.05	0.000674	0.004639
pet_read		AB+	0.2705	0.03316	154	8.16	<.0001	0.05	0.2050	0.3360
pet_read		AB-	0	.	.	.	.	.	.	.
Network	DAT		-0.02377	0.007477	154	-3.18	0.0018	0.05	-0.03854	-0.00900
Network	DMN		-0.01315	0.008999	154	-1.46	0.1461	0.05	-0.03092	0.004631
Network	FPN		-0.01386	0.009860	154	-1.41	0.1619	0.05	-0.03334	0.005621
Network	LIM		0.008669	0.008329	154	1.04	0.2996	0.05	-0.00778	0.02512
Network	SOM		-0.07266	0.007981	154	-9.10	<.0001	0.05	-0.08843	-0.05690
Network	VAT		0.001538	0.009461	154	0.16	0.8711	0.05	-0.01715	0.02023
Network	VIS		0	.	.	.	.	.	.	.
Network*pet_read	DAT	AB+	0.1162	0.01770	154	6.56	<.0001	0.05	0.08124	0.1512
Network*pet_read	DAT	AB-	0	.	.	.	.	.	.	.
Network*pet_read	DMN	AB+	0.1586	0.02131	154	7.44	<.0001	0.05	0.1165	0.2007
Network*pet_read	DMN	AB-	0	.	.	.	.	.	.	.
Network*pet_read	FPN	AB+	0.1780	0.02335	154	7.62	<.0001	0.05	0.1318	0.2241
Network*pet_read	FPN	AB-	0	.	.	.	.	.	.	.

Solution for Fixed Effects										
Effect	Network	pet_read	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
Network*pet_read	LIM	AB+	0.000588	0.01972	154	0.03	0.9762	0.05	-0.03837	0.03955
Network*pet_read	LIM	AB-	0	.	.	.	.	.	.	.
Network*pet_read	SOM	AB+	-0.01744	0.01890	154	-0.92	0.3575	0.05	-0.05478	0.01989
Network*pet_read	SOM	AB-	0	.	.	.	.	.	.	.
Network*pet_read	VAT	AB+	0.1078	0.02240	154	4.81	<.0001	0.05	0.06358	0.1521
Network*pet_read	VAT	AB-	0	.	.	.	.	.	.	.
Network*pet_read	VIS	AB+	0	.	.	.	.	.	.	.
Network*pet_read	VIS	AB-	0	.	.	.	.	.	.	.

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
age	1	154	7.01	0.0089
pet_read	1	154	114.87	<.0001
Network	6	154	88.37	<.0001
Network*pet_read	6	154	29.53	<.0001

Least Squares Means										
Effect	Network	pet_read	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
Network*pet_read	DAT	AB+	1.5610	0.03002	154	52.01	<.0001	0.05	1.5017	1.6203
Network*pet_read	DAT	AB-	1.1743	0.01388	154	84.60	<.0001	0.05	1.1469	1.2017
Network*pet_read	DMN	AB+	1.6140	0.03427	154	47.09	<.0001	0.05	1.5462	1.6817
Network*pet_read	DMN	AB-	1.1849	0.01588	154	74.63	<.0001	0.05	1.1535	1.2163
Network*pet_read	FPN	AB+	1.6326	0.03406	154	47.93	<.0001	0.05	1.5654	1.6999
Network*pet_read	FPN	AB-	1.1842	0.01578	154	75.05	<.0001	0.05	1.1530	1.2154
Network*pet_read	LIM	AB+	1.4778	0.03240	154	45.61	<.0001	0.05	1.4138	1.5418
Network*pet_read	LIM	AB-	1.2067	0.01500	154	80.45	<.0001	0.05	1.1771	1.2363
Network*pet_read	SOM	AB+	1.3784	0.02498	154	55.18	<.0001	0.05	1.3291	1.4278
Network*pet_read	SOM	AB-	1.1254	0.01151	154	97.74	<.0001	0.05	1.1026	1.1481
Network*pet_read	VAT	AB+	1.5779	0.03084	154	51.17	<.0001	0.05	1.5170	1.6388
Network*pet_read	VAT	AB-	1.1996	0.01427	154	84.09	<.0001	0.05	1.1714	1.2278
Network*pet_read	VIS	AB+	1.4685	0.02999	154	48.96	<.0001	0.05	1.4093	1.5278
Network*pet_read	VIS	AB-	1.1980	0.01387	154	86.37	<.0001	0.05	1.1706	1.2254